PARIKSHA365 — STUDY NOTES

Computer Proficiency — for SSC CGL Tier-2 Paper-I (Section III Module-I)

> Computer Basics + MS Office + Internet & E-mail + Networking & Cyber Security

SSC

v 2026-04

The promise. Read this book end-to-end (with the usual two re-reads + active recall on the embedded prompts) and you will be able to attempt every quiz question in the Pariksha365 pool for this subject with a strong fundamental grasp.

How to use this book

Tier-2 Paper-I Section III Module-I is **20 Qs × 3 marks = 60 marks** and is **qualifying-only**, but a cut-off of **40 % (24 marks → about 8 correct)** is mandatory for category-wise selection. Most aspirants neglect this section, then panic on exam day because every term is unfamiliar. This book is engineered so that **40 hours of focused reading + drill** locks the qualifying marks.

The book follows the SSC CGL Tier-2 Section III Module-I syllabus exactly:

Module	Coverage	Pages	Hours
1. Computer Basics	CPU, I/O, memory, storage, ports, Windows Explorer, shortcuts	~14	6
2. Software & MS Office	Windows OS, MS Word, Excel, PowerPoint — features + shortcuts	~22	12
3. Internet & E-mail	Browsers, search, downloads, email mgmt, e-banking	~10	6
4. Networking & Cyber Security	Devices, protocols, threats (virus/worm/trojan), prevention	~14	8
5. PYQ-style Mock + Glossary	50 solved Qs + 200-term glossary	~10	8

Read order: Module 1 → Module 2 (Excel is the most-tested) → Module 4 (cyber-security is rising every year) → Module 3 → Drill the mock.

Tip: Computer Knowledge is **fact-recall heavy** with very little reasoning. Use flash-cards for shortcuts (MS Word/Excel) and acronyms (HTTP, FTP, IMAP, etc.). Active-recall after each chapter, not after the whole book.

Paper map and importance dashboard

Sub-topic	Avg Qs in last 4 Tier-2 papers	Difficulty	Priority
MS Excel — formulas, shortcuts, functions	4-5	Medium	CRITICAL
MS Word — formatting, shortcuts	2-3	Easy	CRITICAL
Computer hardware (CPU/RAM/Memory)	2-3	Easy	CRITICAL
Networking devices & protocols	2-3	Medium	HIGH
Cyber-security threats	2	Easy-Medium	HIGH
Internet (browsers, search, URL)	1-2	Easy	HIGH
MS PowerPoint	1	Easy	MEDIUM
Email & e-banking	1	Easy	MEDIUM
Windows OS / Explorer	1-2	Easy	MEDIUM
File extensions & generations	1	Easy	LOW
Database basics (DBMS, SQL, ACID)	1-2	Easy-Medium	MEDIUM

Pareto rule for 24/60: master Excel shortcuts + hardware basics + cyber-security acronyms. That alone yields 18-20 marks consistently.

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PART 0 — FOUNDATIONS YOU CANNOT SKIP

0.1 What is a computer? (one-line definition the exam asks)

A **computer** is an **electronic device** that **accepts input, processes it according to stored instructions (programs), and produces output**. Five characteristics: **speed, accuracy, diligence, versatility, storage**. Limitations: **no thinking, no IQ, no decision-making without a program, no feelings**.

0.2 Generations of computers (1 Q every other paper)

Generation	Years	Switching technology	Example	Language
1st	1940-56	Vacuum tubes	ENIAC, UNIVAC	Machine language
2nd	1956-63	Transistors	IBM 1401, IBM 7090	Assembly, FORTRAN, COBOL
3rd	1964-71	Integrated Circuits (IC)	IBM 360, ICL 1900	High-level (C, PL/1)
4th	1971-present	Microprocessors (VLSI)	IBM PC, Apple II	C++, Java
5th	Present-future	AI, parallel processing, ULSI	Quantum computers	Prolog, LISP

Memory peg: **Vacuum → Transistor → IC → Microprocessor → AI** = "Very Talented Indian Marathon-Athlete".

0.3 Classification of computers (by purpose / size)

By purpose:

Type	How it works	Real-world examples
Analog	Measures *continuous* physical quantities	Speedometer, thermometer, fuel gauge
Digital	Operates on binary 0/1 (discrete values)	PC, laptop, smartphone — everything we use today
Hybrid	Combines analog sensing + digital processing	ECG machine, petrol-pump display

By size (largest → smallest):

Type	Speed / Cost	Typical use	Examples
Supercomputer	Fastest, most expensive; petaFLOPS range	Weather forecasting, nuclear simulation, AI research	**El Capitan** (USA, 2024 world #1); India: **AIRAWAT** (C-DAC Pune), **PARAM-Siddhi AI** (C-DAC), **Pratyush / Mihir** (NCMRWF — weather)
Mainframe	Very fast, highly reliable	Banks, railways, government (millions of transactions)	IBM Z-series; IRCTC backend
Mini / Server	Mid-range	Mid-sized company servers	IBM AS/400 series
Microcomputer	Affordable, personal use	PC, laptop, tablet, smartphone	Desktop, MacBook, iPad
Workstation	High-end microcomputer	CAD/CAM, 3D rendering, engineering design	Dell Precision, HP Z-series

0.4 Number systems (foundation for everything digital)

System	Base	Digits used	Example
Binary	2	0, 1	1011
Octal	8	0-7	173
Decimal	10	0-9	156
Hexadecimal	16	0-9, A-F	F4

Conversion drill (each appears 1 Q per paper):

1. **Decimal → Binary**: divide by 2 repeatedly, write remainders bottom-up. - 25 → 25/2=12 R1 | 12/2=6 R0 | 6/2=3 R0 | 3/2=1 R1 | 1/2=0 R1 → **11001**.
2. **Binary → Decimal**: multiply each bit by its place-value 2^n. - 11001 → 1×16 + 1×8 + 0×4 + 0×2 + 1×1 = **25**.
3. **Decimal → Hexadecimal**: divide by 16 repeatedly. - 255 → 255/16 = 15 R15 → F & F → **FF**.
4. **Binary → Hex**: group binary into 4 bits from the right. - 11111111 → 1111 1111 → **FF**.
5. **Binary → Octal**: group binary into 3 bits from the right. - 1011010 → 1 011 010 → **132**.

Bit, Byte, etc.: - 1 bit = 1 binary digit (0 or 1). - 1 nibble = 4 bits. - 1 byte = 8 bits. - 1 KB = 1024 bytes (technically; many use 1000). - 1 MB = 1024 KB; 1 GB = 1024 MB; 1 TB = 1024 GB; 1 PB = 1024 TB.

ASCII = American Standard Code for Information Interchange — 7-bit (128 chars) or 8-bit extended (256 chars). **Unicode** = 16-bit (65,536 chars) — supports all languages including Devanagari.

CHAPTER 1 — Computer Basics & Hardware

1.0 The big picture — the von Neumann architecture

Every modern computer obeys the **von Neumann (or stored-program) model**: programs and data live in the same memory; the CPU fetches an instruction, decodes it, executes it, then fetches the next one. Five functional units form the backbone:

```
┌─────────┐       ┌─────────────────────────┐       ┌─────────┐
│  Input  │ ────► │   CPU                   │ ────► │ Output  │
│ devices │       │   ┌────┐ ┌────┐ ┌────┐  │       │ devices │
└─────────┘       │   │ALU │ │ CU │ │REG │  │       └─────────┘
                  │   └────┘ └────┘ └────┘  │
                  └────────────▲────────────┘
                               │
                       ┌───────▼──────┐
                       │   Memory     │  (RAM + ROM + Storage)
                       └──────────────┘
```

The Central Processing Unit has three internal blocks: 1. **ALU** — Arithmetic Logic Unit (does + − × ÷ AND OR NOT comparisons). 2. **CU** — Control Unit (fetches, decodes, dispatches instructions; the "traffic cop"). 3. **Registers** — tiny ultra-fast memory inside the CPU (Accumulator, Program Counter, Instruction Register, MAR, MBR).

Most-tested fact: the CPU is called the **brain of the computer**. Its speed is measured in **clock cycles per second** — **Hertz (Hz)**, today GigaHertz (GHz).

1.1 Input devices — full taxonomy

Device	Purpose	Notes
Keyboard	Text input	Standard layout = QWERTY (Sholes, 1873)
Mouse	Pointing	Invented by **Douglas Engelbart, 1964**
Trackball / Touchpad / Joystick	Pointing (laptops, gaming)	
Light pen	Direct screen pointing	Pre-touchscreen
Touch screen	Direct touch	Capacitive (smartphones), Resistive (ATM, POS)
Scanner	Convert paper image → digital file	Flatbed / handheld / drum
OCR	Optical Character Recognition — image to editable text	Used by India Post for PIN sorting
OMR	Optical Mark Recognition — reads pencil marks	Used by SSC/UPSC for answer sheets!
MICR	Magnetic Ink Character Recognition — reads cheque numbers	Bank cheque processing
Barcode reader	Reads 1D bar pattern (UPC, EAN)	Retail POS
QR scanner	Reads 2D Quick-Response codes	UPI payment apps
Biometric scanner	Fingerprint, iris, face	Aadhaar, e-passport
Microphone	Voice input	Speech-to-text
Webcam / digital camera	Image / video input	
Stylus / digitiser pad	Drawing tablet (Wacom)	Artists

Two SSC-favourites: - **OMR is what your own answer-sheet uses** — magnetic ink is NOT involved; that's MICR (cheques). - **MICR is a 9-digit code** printed at the bottom of cheques: first 3 digits = city, next 3 = bank, last 3 = branch.

1.2 Output devices — full taxonomy

Device	Type	Notes
Monitor (VDU)	Soft copy	CRT (old) → LCD → LED → OLED → QLED
Printer (Impact)	Hard copy	Dot-matrix, Daisy-wheel — uses ribbon, noisy, cheap
Printer (Non-impact)	Hard copy	Inkjet (sprays ink), Laser (toner + drum), Thermal (receipts)
Plotter	Large drawings	Used in CAD, engineering, banners
Speakers / Headphones	Audio output	
Projector	Large-screen output	LCD/DLP/LCoS
VR Headset	3-D immersive output	Oculus, HTC Vive

Printer resolution is measured in **DPI** (dots per inch); higher = sharper. Monitor resolution is in **pixels** (e.g., 1920 × 1080 = Full HD); pixel density is **PPI**.

Most-asked exam fact: Laser printers are non-impact and use a TONER + photosensitive drum + heated fuser, not ink.

1.3 The motherboard — the central circuit

The motherboard hosts the CPU socket, RAM slots, expansion slots (PCIe), chipset, BIOS chip, SATA / NVMe connectors, and ports. Key chips:

- **Northbridge** (old): connected CPU to RAM and graphics card. In modern designs, the memory controller is inside the CPU.
- **Southbridge / PCH** (Platform Controller Hub): manages slower peripherals (USB, audio, SATA, network).
- **BIOS / UEFI** (chip on motherboard): firmware that boots the OS; UEFI replaces legacy BIOS, supports >2 TB disks, GUI, secure boot.
- **CMOS battery** (CR2032 coin cell): preserves BIOS settings + system clock when computer is off.
- **POST** (Power-On Self-Test): checks RAM/CPU/keyboard at boot; beep codes signal errors.

1.4 Ports — name, shape, purpose

Port	Shape	Purpose	Generation
PS/2	Round, 6-pin (green = mouse, purple = keyboard)	Legacy keyboard / mouse	Deprecated
VGA	15-pin trapezoid (blue)	Analog video	Old monitors
DVI	Rectangular white connector	Digital video	Old monitors
HDMI	Trapezoidal slim	Digital A/V	TVs, modern monitors
Display Port	Notched, slim	Digital A/V (higher res than HDMI)	Modern monitors
USB-A	Flat rectangle	Universal device	USB 1.0 (12 Mbps) → 2.0 (480 Mbps) → 3.0 (5 Gbps) → 3.2 (20 Gbps) → 4 (40 Gbps)
USB-C	Small reversible oval	Modern universal port (data + video + power)	USB 3.1+
RJ-45	Like phone jack but wider	Ethernet (LAN)	100M / 1G / 10G
RJ-11	Small phone jack	Telephone / DSL	
3.5 mm jack	Round audio	Headphones, mic	

Port	Shape	Purpose	Generation
eSATA	Like internal SATA but external	External hard disk (rare now)	
Thunderbolt	Same shape as USB-C	High-speed (up to 80 Gbps Thunderbolt 5)	Apple + modern laptops

Exam favourite: USB stands for **Universal Serial Bus**.

1.5 Inside the CPU — registers and clock

A CPU contains specialised **registers**:

Register	Function
PC — Program Counter	Holds address of NEXT instruction
IR — Instruction Register	Holds the CURRENT instruction being executed
MAR — Memory Address Register	Address of the memory location being accessed
MBR/MDR — Memory Buffer/Data Register	Data being transferred to/from memory
Accumulator	Stores intermediate ALU results
Stack Pointer	Top of the program stack
Status / Flags	Z (zero), C (carry), N (negative), O (overflow) flags

The **machine cycle** has 4 steps: **Fetch → Decode → Execute → Store** (sometimes labelled "Fetch-Decode-Execute" — the FDE cycle).

Clock speed: how many cycles per second the CPU performs. 3 GHz = 3 billion cycles/second. Modern CPUs use **multi-core** (2, 4, 6, 8, 16 cores), **hyper-threading** (2 logical threads per core), and **cache** (L1 fastest, L2, L3 largest).

> ***Exam tip***: "***Throughput***" = *total useful work per unit time;* "***Response time***" = *time between request and reply. They are different metrics.*

1.6 Worked PYQ-style examples — Hardware

Q1.1 (Easy). The brain of a computer is:

(a) RAM (b) ROM (c) **CPU** (d) Monitor

Solution: CPU performs all processing → (c).

Q1.2 (Easy). Which is a non-impact printer?

(a) Dot-matrix (b) Daisy-wheel (c) **Laser** (d) Drum

Solution: Laser uses toner + drum, no mechanical strike → (c).

Q1.3 (Medium). OMR is used to:

(a) Read magnetic ink on cheques
(b) **Read pencil marks on a printed sheet**
(c) Read bar-codes
(d) Read fingerprints

Solution: OMR = Optical Mark Recognition → (b). Cheques use MICR.

Q1.4 (Medium). Which port is reversible?

(a) USB-A (b) **USB-C** (c) HDMI (d) VGA

Solution: USB-C is symmetric, can be plugged either way → (b).

Q1.5 (Hard). The Program Counter (PC) holds:
(a) Current instruction
(b) **Address of the next instruction**
(c) ALU result
(d) Data being read from memory

Solution: PC = address of NEXT instruction → (b). IR holds the current instruction.

CHAPTER 2 — Computer Memory & Storage

2.0 Memory hierarchy — picture this pyramid

```
              ▲  fastest, smallest, costliest
              |
       ┌──────┴──────┐
       │  Registers  │   (inside CPU, ns)
       ├─────────────┤
       │  Cache L1/2/3│   (on CPU die)
       ├─────────────┤
       │  RAM        │   (main memory, GB-scale, volatile)
       ├─────────────┤
       │  SSD / NVMe │   (fast secondary)
       ├─────────────┤
       │  HDD        │   (mechanical secondary)
       ├─────────────┤
       │  Optical /  │   (CD/DVD/Blu-ray, USB stick)
       │  Removable  │
       ├─────────────┤
       │  Cloud / Tape│   (cheapest, slowest, archival)
       └─────────────┘
              |
              ▼  slowest, largest, cheapest
```

2.1 Primary memory (volatile + non-volatile)

RAM — Random Access Memory. **Volatile** (loses data on power off). Two main types: - **DRAM** (Dynamic) — needs constant refresh; cheap; used as main system RAM. Variants: SDRAM → DDR → DDR2 → DDR3 → DDR4 → DDR5. - **SRAM** (Static) — no refresh, very fast, expensive; used as **CPU cache**.

EXAM ALERT

⚠️ **TRAP — RAM is volatile; ROM is non-volatile:** This is the single most tested memory fact. **RAM** loses all data the instant power is cut (volatile). **ROM** retains its contents without power (non-volatile). A common wrong option says "RAM is permanent storage" — it is NOT. Pen drives, SSDs, and hard disks are also non-volatile (they use Flash or magnetic storage, not RAM).

KEY FACT

⚡ **SHORTCUT — Memory size: 1 KB = 1024 bytes, NOT 1000:** Computers use binary (base-2), so 1 KB = 2^{10} = **1024** bytes. This surprises many students who expect 1000. The chain: 1 KB = 1024 B → 1 MB = 1024 KB → 1 GB = 1024 MB → 1 TB = 1024 GB. (Hard-drive manufacturers deceptively use 1 KB = 1000 B, which is why a "500 GB" drive shows ~465 GB in Windows — examiners test this.)

ROM — Read-Only Memory. **Non-volatile**. Stores firmware (BIOS, embedded code). | Type | Full form | How it's erased | Use | |---|---|---|---| | **PROM** | Programmable ROM | Cannot — write once | Factory firmware | | **EPROM** | Erasable PROM | UV light | Older microcontrollers | | **EEPROM** | Electrically Erasable PROM | Electrical signal | BIOS chips | | **Flash** | (modern EEPROM) | Electrical (block-level) | Pen-drives, SSDs, BIOS |

Cache memory — small SRAM between CPU and RAM. **L1** (smallest, fastest, per-core) → **L2** (larger, per-core or per-cluster) → **L3** (shared across cores).

Virtual memory — disk space used as if it were RAM (Windows: pagefile.sys; Linux: swap partition). Lets you run programs larger than physical RAM but slower.

2.2 Secondary storage devices

Medium	Capacity	Speed	Cost	Notes
HDD (Hard Disk Drive)	500 GB – 20 TB	100 MB/s	Cheap	Mechanical: platters + read-write head; SATA interface
SSD (Solid State Drive)	128 GB – 8 TB	500 MB/s	Medium	NAND flash; SATA or NVMe; no moving parts
NVMe SSD	256 GB – 8 TB	3-7 GB/s	High	Connects via PCIe; ~10× faster than SATA SSD
USB Flash Drive ("pen drive")	4 GB – 1 TB	50-200 MB/s	Cheap	Hot-pluggable
SD card / micro-SD	4 GB – 1 TB	50-300 MB/s	Cheap	Cameras, phones
CD (Compact Disc)	**700 MB**	24× = 3.6 MB/s	Cheap	Read by red laser (780 nm)
DVD (Digital Versatile Disc)	**4.7 GB** (single layer) / 8.5 GB (dual)	16× = 22 MB/s	Cheap	Red laser (650 nm)
Blu-ray	**25 GB** / 50 GB (dual) / 100 GB (BDXL)	36 MB/s	Med	**Blue** laser (405 nm) — hence the name
Magnetic tape	12 TB+	Slow	Very cheap	Sequential access, archival backup

Most-tested numerical facts: | Medium / Interface | Capacity / Evolution | |---|---| | CD | **700 MB** | | DVD (single layer) | **4.7 GB** (≈ 7 CDs) | | Blu-ray (single layer) | **25 GB** (≈ 5 DVDs) | | HDD interfaces | PATA (old) → SATA (current) → SAS (servers) | | SSD interfaces | SATA (2.5") → mSATA → M.2 → NVMe/PCIe |

RAID (Redundant Array of Independent Disks): | RAID | Technique | Redundancy | Min disks | |---|---|---|---| | **RAID 0** | Striping | None (pure speed) | 2 | | **RAID 1** | Mirroring | Full (half capacity) | 2 | | **RAID 5** | Striping + distributed parity | 1-disk failure | 3 | | **RAID 10** | RAID 1 + RAID 0 | High speed + redundancy | 4 |

2.3 File types & extensions (1 Q per paper)

Extension	Meaning
`.docx` / `.doc`	MS Word

Extension	Meaning
`.xlsx` / `.xls`	MS Excel
`.pptx` / `.ppt`	MS PowerPoint
`.pdf`	Portable Document Format (Adobe)
`.txt`	Plain text
`.rtf`	Rich Text Format
`.csv`	Comma-Separated Values
`.html` / `.htm`	Web page
`.css`	Cascading Style Sheet
`.js`	JavaScript
`.exe`	Executable (Windows)
`.bat`	Batch file (Windows)
`.zip` / `.rar` / `.7z`	Compressed archive
`.iso`	Disk image
`.mp3` / `.wav` / `.aac`	Audio
`.mp4` / `.avi` / `.mov` / `.mkv`	Video
`.jpg` / `.png` / `.gif` / `.bmp` / `.svg`	Image
`.psd`	Photoshop
`.tmp`	Temporary file
`.sys`	System file
`.dll`	Dynamic Link Library
`.log`	Log file

2.4 Worked examples — Memory

Q2.1. Which memory is volatile?

(a) ROM (b) **RAM** (c) Hard disk (d) Flash

Solution: RAM loses content on power off → (b).

Q2.2. L1 cache is located:

(a) On the motherboard (b) **Inside the CPU**

(c) On the hard disk (d) In the BIOS chip

Solution: L1 cache is on the CPU die, closest to registers → (b).

Q2.3. A standard single-layer DVD holds:

(a) 700 MB (b) **4.7 GB** (c) 25 GB (d) 1 TB

Solution: Single-layer DVD = 4.7 GB → (b).

Q2.4. Which is non-volatile?

(a) RAM (b) Cache (c) Register (d) **ROM**

Solution: ROM retains contents without power → (d).

Q2.5. Pen-drive memory technology is:

(a) DRAM (b) SRAM

(c) **Flash (EEPROM)** (d) HDD

Solution: USB flash drives use NAND flash, a form of EEPROM → (c).

CHAPTER 3 — Operating System & Windows Explorer

3.0 What does an OS do?

The Operating System is the **bridge between hardware and the user**. Five core jobs:

1. **Process management** — schedules CPU among processes (round-robin, priority).
2. **Memory management** — allocates RAM, handles virtual memory & paging.
3. **File system** — organises files in folders, controls permissions.
4. **Device management** — drivers + I/O queues for printers, disks, USB.
5. **Security** — user accounts, passwords, access control lists.

Popular OS families: | OS family | Developer | Type | Versions / distros | |---|---|---|---| | **Windows** | Microsoft | Proprietary GUI | 95, XP, 7, 10, 11; Server 2016/19/22 | | **macOS** | Apple | BSD-Unix based | Big Sur, Monterey, Ventura, Sonoma | | **Linux** | Open-source | Unix-like | Ubuntu, Fedora, Debian, CentOS, Red Hat, Mint, Arch | | **Unix** | Bell Labs (1969) | Original commercial | The ancestor of macOS & Linux | | **Mobile** | — | Mobile OS | Android (Google, Linux kernel), iOS (Apple), HarmonyOS (Huawei) |

3.1 Types of operating system (by usage)

Type	Example	Use case
Batch	Old IBM mainframes	Run jobs in batches without user interaction
Time-sharing / Multi-user	Unix	Multiple users share CPU via time slices
Real-time (RTOS)	VxWorks, QNX	Strict deadline (Mars rover, air-traffic, ABS in cars)
Distributed	Hadoop, Plan 9	Cluster of machines acts as one
Embedded	Watch, microwave OS	Small footprint, dedicated hardware
Single-user	DOS, early Win	One user at a time

3.2 Windows OS — must-know features

- **Desktop** = the screen you see after login.
- **Taskbar** = bottom strip with Start menu, search, pinned apps, system tray, clock.
- **Start Menu** = Windows logo at bottom-left.
- **Recycle Bin** = temporary storage for deleted files (can restore unless emptied). **Shift + Delete** bypasses Recycle Bin.
- **My Computer / This PC** = view of drives (C:, D:, etc.).
- **Control Panel / Settings** = system configuration.
- **Task Manager** (Ctrl + Shift + Esc) = view and kill running processes.
- **File Explorer / Windows Explorer** (Win + E) = browse files and folders.
- **Run dialog** (Win + R) = type command to launch program (e.g., `cmd`, `notepad`, `calc`).
- **Command Prompt** (`cmd`) and **PowerShell** = command-line interfaces.

3.3 File-system facts

File System	OS	Max file size	Key feature
FAT32	Universal	**4 GB**	Most compatible; older/small drives
NTFS	Windows-native	16 TB+	Permissions, encryption, journaling
exFAT	Cross-platform	No 4 GB limit	Flash drives, SD cards for HD video
HFS+ / APFS	macOS	8 EB (APFS)	SSD-optimised; snapshots
ext4	Linux (default)	16 TB	Journaling, large volumes

Path naming: `C:\Users\Bhanu\Documents\note.docx` (Windows uses **backslash** `\`); Linux/Mac use **forward slash** `/`.

3.4 Universal Windows keyboard shortcuts (MUST memorise)

KEY FACT

⚡ SHORTCUT — Core 8 keyboard shortcuts every SSC question tests:

Shortcut	Action	Memory hook
Ctrl + C	Copy	**C**opy
Ctrl + V	Paste	**V**irtually puts it back
Ctrl + X	Cut	Sc**X**issors cut
Ctrl + Z	Undo	**Z**ap the last action
Ctrl + A	Select All	**A**ll
Ctrl + S	Save	**S**ave
Ctrl + P	Print	**P**rint
Ctrl + F	Find	**F**ind

Mnemonic for the order: **"Copy it, X-cut it, Zip it, All Save, Print-Find."**

Shortcut	Action
Ctrl + C	Copy
Ctrl + X	Cut
Ctrl + V	Paste
Ctrl + Z	Undo
Ctrl + Y	Redo
Ctrl + A	Select all
Ctrl + S	Save
Ctrl + P	Print
Ctrl + F	Find
Ctrl + H	Find & Replace
Ctrl + N	New file/window
Ctrl + O	Open
Ctrl + W	Close current window/tab
Ctrl + T	New tab (browser)
Alt + Tab	Switch between open windows

Shortcut	Action
Alt + F4	Close current application
Win + D	Show desktop / minimise all
Win + E	Open File Explorer
Win + L	Lock the computer
Win + R	Open Run dialog
Win + Tab	Task view (virtual desktops)
Win + I	Open Settings
Win + S	Open Search
Win + Print Screen	Save screenshot to Pictures/Screenshots
Win + Shift + S	Snipping tool (partial screenshot)
F1	Help
F2	Rename selected file
F5	Refresh
F11	Full screen (browser)
Ctrl + Shift + Esc	Task Manager (direct)
Ctrl + Alt + Del	Security options screen
Print Screen	Capture full screen to clipboard

3.5 Worked examples — OS

Q3.1. The shortcut to lock a Windows computer is:
(a) Win + D (b) **Win + L** (c) Win + E (d) Win + R

Solution: Win + L locks the screen → (b).

Q3.2. Which file system has a 4-GB single-file limit?
(a) NTFS (b) **FAT32** (c) exFAT (d) ext4

Solution: FAT32 single-file max = 4 GB minus 1 byte → (b).

Q3.3. Pressing F5 in File Explorer:
(a) Renames the file (b) Deletes the file
(c) **Refreshes the window** (d) Opens a new folder

Solution: F5 = refresh universally → (c).

Q3.4. Linux is best described as:
(a) A spreadsheet program
(b) **An open-source operating system**
(c) A word processor
(d) An anti-virus

Solution: Linux is an OS, open-source → (b).

CHAPTER 4 — MS Word

4.0 What is MS Word?

MS Word is a **word-processing software** in the **Microsoft Office suite**, first released in 1983 (then Multi-Tool Word for Xenix). Current versions: **Word 2016 / 2019 / 2021 / Microsoft 365**. Default file format since 2007: `.docx` (Office Open XML); older: `.doc`.

KEY FACT

Microsoft 365 vs Office 2021 — the difference: - **Office 2021 (perpetual):** Buy once, own forever; fixed features at time of purchase; no cloud sync by default. - **Microsoft 365 (subscription):** Monthly/annual subscription; always has the latest features; includes OneDrive cloud storage, Teams, and online apps. This is Microsoft's primary offering from 2022 onwards for most users. Exam tip: When a question asks about the "current" or "cloud-based" Office suite, **Microsoft 365** is the answer.

When you open Word, you see the **Ribbon** (introduced in Word 2007) replacing menus. The Ribbon has **tabs**: File, Home, Insert, Draw, Design, Layout, References, Mailings, Review, View, Help.

4.1 Anatomy of a Word window

```
┌──────────────────────────────────────────────────────────────┐
│  Title bar — "Document1 - Word"      [_] [□] [×]            │
├──────────────────────────────────────────────────────────────┤
│  File  Home  Insert  Design  Layout  References  …           │  ← Ribbon tabs
├──────────────────────────────────────────────────────────────┤
│  [Paste] [B I U] [Font: Calibri ▼] [11▼] [Align] [Styles]    │  ← Ribbon contents
├──────────────────────────────────────────────────────────────┤
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├──────────────────────────────────────────────────────────────┤
│                                                              │
│                                                              │
│                  D O C U M E N T   A R E A                   │
│                                                              │
│                                                              │
├──────────────────────────────────────────────────────────────┤
│ Page 1 of 1 │ 254 words │ ✔ │      [layout]  100% [- + ] │  ← Status bar + Zoom
└──────────────────────────────────────────────────────────────┘
```

4.2 Ribbon tabs — key features at a glance

Tab	Key features
Home	Clipboard, Font, Paragraph, Styles, Editing (Find/Replace)
Insert	Pages (cover, blank, break), Tables, Pictures, Shapes, SmartArt, Chart, Links (hyperlink), Header/Footer, Page Number, Text Box, WordArt, Equation, Symbol
Design	Document Themes, Page Color, Watermark, Page Borders
Layout	Margins, Orientation (Portrait/Landscape), Size, Columns, Breaks, Line Numbers, Hyphenation, Spacing
References	Table of Contents, Footnotes, Citations & Bibliography, Captions, Index
Mailings	Mail Merge (start, select recipients, write & insert fields, preview, finish & merge)

Tab	Key features
Review	Spelling & Grammar, Thesaurus, Word Count, Translate, Comments, Tracking (Track Changes), Compare, Protect
View	Read Mode / Print Layout / Web Layout / Outline / Draft, Ruler, Gridlines, Zoom, Window arrangement

4.3 Important features explained

Mail Merge — combines a main document (letter / envelope / label) with a data source (Excel/Access/CSV) to produce personalised copies. Steps: Mailings tab → Start Mail Merge → Select Recipients → Insert Merge Field → Preview → Finish & Merge.

Track Changes — Review tab → Track Changes (Ctrl + Shift + E). Every edit shows in red; accept / reject via Review → Accept / Reject.

Header & Footer — Insert tab → Header / Footer. Page numbers, document title, date. Different first page / different odd-even possible.

Page Break vs Section Break — - **Page Break** (Ctrl + Enter) — moves to next page; formatting unchanged. - **Section Break** — splits document so each section can have its own orientation, margins, header/footer, page numbers.

Styles — pre-defined formatting (Heading 1, Heading 2, Normal, Quote, Title). Use styles consistently so Table of Contents updates automatically.

Table of Contents — References → Table of Contents → choose Automatic Table 1/2. Update by right-click → Update Field.

Hyperlink (Ctrl + K) — Insert tab → Link → URL or another location.

Footnote / Endnote — References → Insert Footnote (Alt + Ctrl + F) / Insert Endnote.

Find & Replace — Ctrl + H. Supports wildcards, case-match, whole-word.

Word Count — Review → Word Count (or status bar at bottom).

Save vs Save As — Ctrl + S saves; **F12** = Save As.

4.4 Word-specific keyboard shortcuts (the high-yield 30)

Shortcut	Action
Ctrl + B	Bold
Ctrl + I	Italic
Ctrl + U	Underline
Ctrl + E	Centre align
Ctrl + L	Left align
Ctrl + R	Right align
Ctrl + J	Justify
Ctrl + 1	Single spacing
Ctrl + 2	Double spacing
Ctrl + 5	1.5 line spacing
Ctrl + Shift + >	Increase font size
Ctrl + Shift + <	Decrease font size
Ctrl +]	Increase by 1 pt
Ctrl + [	Decrease by 1 pt

Shortcut	Action
Ctrl + Shift + D	Double underline
Ctrl + Shift + K	Small caps
Ctrl + =	Subscript
Ctrl + Shift + =	Superscript
Ctrl + Backspace	Delete previous word
Ctrl + Delete	Delete next word
Ctrl + Home	Top of document
Ctrl + End	End of document
Ctrl + Enter	Page break
Shift + Enter	Line break (no new paragraph)
Ctrl + K	Insert hyperlink
Alt + Ctrl + F	Insert footnote
F7	Spell check
Shift + F7	Thesaurus
F12	Save As
Ctrl + Shift + Spacebar	Non-breaking space

4.5 Worked examples — Word

Q4.1. Default file extension for MS Word 2019:

(a) .doc (b) **.docx** (c) .txt (d) .rtf

Solution: Since Word 2007, default = .docx → (b).

Q4.2. Shortcut to **make text bold**:

(a) Ctrl + I (b) Ctrl + U (c) **Ctrl + B** (d) Ctrl + E

Solution: B = Bold → (c).

Q4.3. Which tab contains "Mail Merge"?

(a) Home (b) Insert (c) **Mailings** (d) Review

Solution: Mailings tab → (c).

Q4.4. Shortcut for "Save As" is:

(a) Ctrl + S (b) **F12** (c) Alt + F12 (d) Shift + F12

Solution: F12 opens Save As dialog → (b).

Q4.5. A **section break** is used to:

(a) Start a new page

(b) **Allow different page formatting in different parts of the document**

(c) Insert a footnote

(d) Hyphenate words

Solution: Section break lets each section have different orientation/margins/header → (b).

CHAPTER 5 — MS Excel

5.0 Why Excel is the most-tested topic

Excel appears in roughly **4-5 of 20 Qs** in the Tier-2 computer section — that is **~25 %** of the section weight. Most questions test: - Default file extension (.xlsx) - Cell-reference types (relative vs absolute vs mixed: A1, $A1$,A1, A1) - Common functions (SUM, AVERAGE, COUNT, IF, VLOOKUP, COUNTIF, SUMIF, NOW, TODAY, CONCATENATE, LEN, LEFT, RIGHT, MID, ROUND, MAX, MIN) - Shortcuts (Ctrl + ; for today's date, F4 to toggle absolute reference, Alt + = for autosum) - Chart types (column, bar, line, pie, scatter, area) - Pivot tables

5.1 Excel anatomy

A workbook (.xlsx file) contains **worksheets** (tabs at bottom). Each worksheet is a grid of **rows** (numbered 1, 2, 3...) and **columns** (lettered A, B, C ... Z, AA, AB ... ZZ ... XFD). The intersection of a row and column is a **cell**, addressed like **B7** or B**7**.

Limits (Excel 2016+): **1,048,576 rows × 16,384 columns** per worksheet. Max worksheets per workbook: limited by RAM.

The **Name Box** (top-left) shows the address of the active cell. The **Formula Bar** to its right shows the cell's content/formula. The active cell has a thick border with a **fill handle** (small square at bottom-right corner) for dragging formulas.

5.2 Cell references — the make-or-break concept

Every formula uses references. There are **three reference types**:

Reference	Example	Behaviour when copied
Relative	`A1`	Both row & column shift with the copy
Absolute	`$A$1`	Neither row nor column shifts (locked)
Mixed (column locked)	`$A1`	Column stays; row shifts
Mixed (row locked)	`A$1`	Row stays; column shifts

Toggle with F4: select the reference inside a formula and press F4 repeatedly to cycle `A1 → $A$1 → A$1 → $A1 → A1`.

Why it matters: in the formula `=A1$*B$1` copied from C1 down to C5, A1 becomes A2, A3, A4, A5 — but B1 stays fixed (e.g., a multiplier or tax rate).

5.3 The 30 most-asked functions

Function	Purpose	Example
SUM	Add	`=SUM(A1:A10)`
AVERAGE	Mean	`=AVERAGE(A1:A10)`
MIN / MAX	Smallest / largest	`=MAX(A1:A10)`
COUNT	Count cells with numbers	`=COUNT(A1:A10)`
COUNTA	Count non-blank cells	`=COUNTA(A1:A10)`
COUNTBLANK	Count blanks	`=COUNTBLANK(A1:A10)`
COUNTIF	Count cells meeting condition	`=COUNTIF(A1:A10,">50")`
SUMIF	Sum cells meeting condition	`=SUMIF(A1:A10,">50")`

Function	Purpose	Example
SUMIFS	Sum with multiple conditions	`=SUMIFS(C1:C10,A1:A10,"Apple",B1:B10,">100")`
IF	Conditional	`=IF(A1>50,"Pass","Fail")`
IFS (2019+)	Multiple IF	`=IFS(A1≥90,"A",A1≥80,"B",A1≥60,"C",TRUE,"F")`
AND / OR / NOT	Logical	`=IF(AND(A1>0,B1>0),"OK","X")`
VLOOKUP	Look up value in first column, return from another	`=VLOOKUP(D1,A1:B10,2,FALSE)`
HLOOKUP	Same, but horizontal	
XLOOKUP (365)	Modern replacement, can search any direction	
INDEX + MATCH	Powerful 2-D lookup	`=INDEX(B1:B10,MATCH("Sales",A1:A10,0))`
CONCATENATE / CONCAT / &	Join strings	`=A1 & " " & B1`
LEN	String length	`=LEN(A1)`
LEFT / RIGHT / MID	Substring	`=LEFT(A1,3)` , `=MID(A1,4,2)`
UPPER / LOWER / PROPER	Change case	`=PROPER(A1)`
TRIM	Remove extra spaces	`=TRIM(A1)`
SUBSTITUTE	Replace text	`=SUBSTITUTE(A1,"old","new")`
TEXT	Format number as text	`=TEXT(NOW(),"dd-mmm-yyyy")`
VALUE	Convert text to number	`=VALUE("123")`
ROUND / ROUNDUP / ROUNDDOWN	Round number	`=ROUND(A1,2)`
INT / MOD	Integer / modulo	`=MOD(10,3)` → 1
POWER / SQRT	Powers	`=POWER(2,8)` → 256
NOW / TODAY	Current datetime / date	`=NOW()` , `=TODAY()`
DATE / YEAR / MONTH / DAY	Build / extract date parts	`=DATE(2026,4,15)`
DATEDIF	Difference between dates	`=DATEDIF(A1,B1,"Y")` for years
PMT	Loan EMI	`=PMT(rate/12, n_months, -principal)`

5.4 Excel error messages — what each means

Error	Meaning
#DIV/0!	Dividing by zero
#NAME?	Function or named range typo
#VALUE!	Wrong type of argument (text where number expected)
#REF!	Reference to a deleted cell
#N/A	Value not available (VLOOKUP miss)
#NULL!	Wrong range operator (e.g., space instead of comma)
#NUM!	Invalid numeric value (sqrt of negative, too large)
####	Column too narrow to display number (widen column)

5.5 Charts — when to use what

Chart type	Best for
Column	Comparing values across categories
Bar	Same as column but horizontal (long labels)
Line	Trends over time
Pie	Parts of a whole (≤ 6 slices)
Doughnut	Like pie with hole; can compare multiple series
Area	Cumulative value over time
Scatter (XY)	Correlation between two numeric variables
Bubble	3-variable scatter (size = third)
Histogram	Distribution of a single variable
Box & Whisker	Quartiles, outliers
Combo	Two chart types overlaid (e.g., bar + line)
Waterfall	Show running total as additions/subtractions
Treemap	Hierarchical data as rectangles
Sunburst	Hierarchical data as concentric rings

Sparkline = a tiny chart inside one cell (Insert → Sparklines → Line / Column / Win-Loss).

5.6 Excel shortcuts — the high-yield 35

Shortcut	Action
Ctrl + ;	Insert today's date
Ctrl + Shift + ;	Insert current time
Ctrl + A	Select all (entire data region; press again for whole sheet)
Ctrl + Spacebar	Select entire column
Shift + Spacebar	Select entire row
Ctrl + Shift + L	Toggle AutoFilter
Ctrl + T	Convert range to Table
Ctrl + Shift + +	Insert row/column
Ctrl + -	Delete row/column
Alt + =	AutoSum
Ctrl + Enter	Fill selected cells with same value
Ctrl + D	Fill Down
Ctrl + R	Fill Right
Ctrl + Page Down / Page Up	Switch worksheet tabs
Ctrl + N	New workbook
Ctrl + W	Close workbook
Ctrl + Home	Go to A1
Ctrl + End	Go to last used cell
Ctrl + Arrow	Jump to edge of data block
Shift + Arrow	Extend selection one cell
Ctrl + Shift + Arrow	Extend selection to edge of data

Shortcut	Action
F2	Edit active cell
F4	Repeat last action / toggle reference (in formula)
F5 / Ctrl + G	Go To dialog
F9	Recalculate all workbooks
Alt + Enter	New line within a cell
Ctrl + 1	Format Cells dialog
Ctrl + Shift + $	Currency format
Ctrl + Shift + %	Percentage format
Ctrl + Shift + #	Date format
Ctrl + Shift + @	Time format
Ctrl + Shift + !	Number format with thousand separator
Ctrl + B / I / U	Bold / Italic / Underline
Ctrl + 5	Strikethrough
Ctrl + K	Insert hyperlink
Ctrl + Shift + U	Expand/collapse formula bar

5.7 Conditional formatting + pivot tables — what to know

Conditional Formatting (Home → Conditional Formatting) lets cells change colour based on rules (e.g., highlight all cells > 80 in green). Built-in types: **Highlight Cell Rules**, **Top/Bottom Rules**, **Data Bars**, **Color Scales**, **Icon Sets**, **New Rule** (custom formula).

Pivot Table (Insert → PivotTable) summarises large tables by grouping. Drag fields into: - **Rows** — what becomes the row labels - **Columns** — what becomes the column labels - **Values** — what gets aggregated (Sum, Count, Average, etc.) - **Filters** — slicer to limit data Pivot tables auto-refresh on Data → Refresh All (Ctrl + Alt + F5).

5.8 Worked examples — Excel

Q5.1. Default extension of an Excel 2019 workbook:
(a) .xls (b) **.xlsx** (c) .csv (d) .xlsm

Solution: .xlsx is the default since 2007; .xlsm is macro-enabled → (b).

Q5.2. Shortcut to insert today's date in active cell:
(a) Ctrl + T (b) **Ctrl + ;** (c) Ctrl + D (d) Alt + D

Solution: Ctrl + ; for date; Ctrl + Shift + ; for time → (b).

Q5.3. `=SUMIF(A1:A10,">100")` returns:
(a) Sum of A1:A10
(b) **Sum of cells in A1:A10 that are greater than 100**
(c) Count of cells > 100
(d) Average of cells > 100

Solution: SUMIF sums cells meeting the criterion → (b).

Q5.4. Which is an **absolute** cell reference?
(a) A1 (b) $A1$ (c) A1 (d) **A1**

Solution: Both row and column locked with $ → (d).

Q5.5. Error `#DIV/0!` means:
(a) Wrong function name (b) **Division by zero**
(c) Reference deleted (d) Wrong data type

Solution: A cell is being divided by 0 → (b).

Q5.6. Cell B3 contains `=A1+A2`. If you copy this to D5, the formula becomes:
(a) `=A1+A2` (b) `=C3+C4` (c) **`=C3+C4`** (d) `=D5+D6`

Solution: Relative shift: B3→D5 means columns shift right by 2 (B→D) and rows shift down by 2 (3→5). So A1→C3, A2→C4. Answer: `=C3+C4` → (c).

Q5.7. Which function joins text values into one string?
(a) **CONCATENATE** (b) JOIN
(c) MERGE (d) ADD

Solution: CONCATENATE or `&` → (a).

Q5.8. Maximum number of columns in an Excel 2016 worksheet:
(a) 256 (b) 16,384 (c) **16,384** (d) 1,048,576

Solution: 16,384 columns (A to XFD); rows = 1,048,576 → (b/c) = 16,384.

Q5.9. Pressing F4 inside a formula on a selected reference:
(a) Recalculates the formula
(b) Opens Find dialog
(c) **Toggles between relative, absolute, and mixed references**
(d) Inserts a new row

Solution: F4 cycles reference types → (c).

Q5.10. To start a formula in Excel, you begin with:
(a) `:` (b) `+` (c) **`=`** (d) `@`

Solution: All formulas start with `=` (or `+` / `-` which Excel converts to `=`) → (c).

CHAPTER 6 — MS PowerPoint

6.0 Quick orientation

MS PowerPoint is **presentation software** in the MS Office suite. Default extension since 2007: **.pptx** (older: .ppt). A presentation is a collection of **slides**; each slide can hold text, images, tables, charts, audio, video, SmartArt, animations.

6.1 Anatomy

```
┌──────────────────────────────────────────────────────────────┐
│   Ribbon (Home / Insert / Design / Transitions / Animations │
│           / Slide Show / Review / View)                      │
├──────────┬───────────────────────────────────────────────────┤
│ Slide    │                                                   │
│ thumbnails│              Active slide canvas                  │
│ (Slide   │                                                   │
│  pane)   │                                                   │
│          │                                                   │
├──────────┴───────────────────────────────────────────────────┤
│ Notes pane (for speaker notes)                               │
└──────────────────────────────────────────────────────────────┘
```

Views: - **Normal** — default editing view. - **Slide Sorter** — thumbnail grid for reordering / deleting. - **Notes Page** — slide + speaker notes printed area. - **Outline** — text-only outline of all slides. - **Reading View** — full-window without going full screen. - **Slide Show / Presenter View** — full screen for delivering.

6.2 Key features

- **Slide Master** (View → Slide Master) — change a master once, all slides update.
- **Layout** — pre-defined slide structure (Title slide, Title and Content, Two Content, Comparison, Blank, etc.).
- **Themes** — colours, fonts, effects applied across all slides.
- **Transition** — animation between slides (Fade, Wipe, Push, Cube, Morph).
- **Animation** — animation within a slide (Appear, Fade, Zoom, Fly In, Wipe, Spin).
- **Hyperlink** (Ctrl + K) — link to another slide, file, web page, or email.
- **Action Button** — shape that triggers an action (next/previous slide, run program).
- **Hide Slide** — exclude a slide from the show without deleting it.
- **Speaker Notes** — text below the slide, visible only in Presenter View.
- **Rehearse Timings** — record how long each slide takes for self-running shows.
- **Pack & Go / Package for CD** — bundles fonts, media, viewer for sharing.

6.3 PowerPoint shortcuts

Shortcut	Action
Ctrl + M	New slide
Ctrl + D	Duplicate selected slide
F5	Start slideshow from beginning
Shift + F5	Start slideshow from current slide
Esc	End slideshow
B (during slideshow)	Black-out screen
W	White-out screen
N / Spacebar / Right arrow	Next slide / animation
P / Backspace / Left arrow	Previous slide

Shortcut	Action
Number + Enter	Jump to slide number
Ctrl + P (during slideshow)	Turn pointer into pen
Ctrl + L	Laser pointer (during slideshow)
Tab	Move to next placeholder
Ctrl + Shift + >	Increase font size
Ctrl + Shift + <	Decrease font size

6.4 Worked examples — PowerPoint

Q6.1. PowerPoint slideshow starts from beginning with:
(a) F4 (b) **F5** (c) Shift + F5 (d) Ctrl + F5

Solution: F5 from beginning; Shift+F5 from current → (b).

Q6.2. During a presentation, pressing **B**:
(a) Bold-faces text (b) **Blacks-out the screen**
(c) Adds a bookmark (d) Adds a slide

Solution: B = black screen; W = white → (b).

Q6.3. Slide Master is found in which tab?
(a) Home (b) Insert (c) Design (d) **View**

Solution: View → Slide Master → (d).

Q6.4. Default extension for PowerPoint 2019:
(a) .ppt (b) **.pptx** (c) .pps (d) .pptm

Solution: .pptx is default; .pptm is macro-enabled → (b).

CHAPTER 7 — Internet, Browsers & Search

7.0 What the Internet is (the one-line definition)

The **Internet** is a **global network of networks** linking billions of devices using the **TCP/IP** protocol suite. The **World Wide Web (WWW)** is a service running over the Internet that uses **HTTP/HTTPS** to share **hypertext documents (web pages)**. So "Internet" ≠ "Web"; the Web is one application on the Internet (like email, FTP, video calls).

7.1 History bullets

- **1969** — ARPANET (predecessor of the Internet) commissioned by US Department of Defence.
- **1989** — **Tim Berners-Lee** invented the World Wide Web at CERN.
- **1991** — First web page went live.
- **1993** — **Mosaic browser** released; popularised the Web.
- **1995** — Internet reached India (commercial launch by VSNL on 15 Aug 1995).
- **2007** — iPhone redefined mobile Internet.

7.2 Web browsers — list and engines

Browser	Developer	Rendering engine	Notes
Chrome	Google	Blink (fork of WebKit)	Most-used worldwide
Edge	Microsoft	Blink (since 2020)	Default on Windows 10/11
Firefox	Mozilla	Gecko	Open-source
Safari	Apple	WebKit	Default on macOS, iOS
Opera	Opera Software	Blink	Built-in VPN, ad-blocker
Brave	Brave Software	Blink	Privacy-first, ad-blocker built-in
Internet Explorer	Microsoft	Trident	**Deprecated** (June 2022)
UC Browser	UC Web (China)	Blink	Popular in India / SE Asia

7.3 Anatomy of a URL

```
https://www.example.com:443/blog/post-1?id=42&page=2#section3
└──┘    └──────────────┘ └┘ └────────┘ └───────────┘ └──────┘
  scheme    host (domain)  port   path        query string  fragment
```

- **Scheme** — `http` or `https` (s = secure, uses TLS encryption).
- **Host** — server's domain name (resolved to IP via DNS).
- **Port** — number identifying service; default 80 (HTTP) or 443 (HTTPS); usually omitted.
- **Path** — folder/file on the server.
- **Query string** — parameters after `?`, separated by `&`.
- **Fragment** — `#section3` jumps to an anchor on the page (no server request).

7.4 Top-level domains (TLDs)

TLD	Meaning
.com	Commercial
.org	Organisation (often non-profit)
.net	Network
.edu	Educational (US universities)
.gov	Government
.mil	US Military
.int	International organisation
.in / .uk / .us / .jp	Country-code (ccTLD)
.gov.in / .nic.in	Indian Government
.ac.in	Indian academic

7.5 Search engines — features

Engine	Operator(s)
Google	`"exact phrase"`, `-exclude`, `site:domain.com`, `filetype:pdf`, `intitle:`, `inurl:`, `OR`, `AND`, `*` (wildcard), `..` (number range)
Bing	Similar to Google plus `feed:` (RSS), `contains:`

Engine	Operator(s)
DuckDuck Go	Privacy-first; `!bang` syntax (e.g., `!w einstein` → Wikipedia)
Yahoo	Now powered by Bing
Yandex / Baidu	Russia / China leaders

Useful Google operators (1 Q every paper): - `site:gov.in budget 2024` — restrict to a domain. - `"Indian Polity" filetype:pdf` — only PDF results. - `-cricket football` — exclude word. - `define:austerity` — get definition. - `weather Delhi` — instant card.

7.6 Downloading and uploading

- **Download** = bringing a file FROM the Internet TO your device.
- **Upload** = sending a file FROM your device TO a server.
- **Bandwidth** = the data-rate of a connection (Mbps).
- **Latency** = round-trip time (ms).
- **Cookies** = small text files stored by sites to remember login / preferences.
- **Cache** = temporary copy of pages/images stored locally to speed up future visits.
- **Bookmark / Favourite** = saved URL inside the browser.
- **Plug-in / Extension** = add-on that extends browser functionality (ad-blocker, password manager).

7.7 Browser shortcuts (Chrome/Edge/Firefox)

Shortcut	Action
Ctrl + T	New tab
Ctrl + W	Close tab
Ctrl + Shift + T	Re-open last closed tab
Ctrl + Tab	Next tab
Ctrl + Shift + Tab	Previous tab
Ctrl + L / F6	Focus address bar
Ctrl + D	Bookmark current page
Ctrl + Shift + N	New incognito/private window
Ctrl + H	History
Ctrl + J	Downloads
Ctrl + + / -	Zoom in / out
Ctrl + 0	Reset zoom
Ctrl + R / F5	Reload
Ctrl + F5	Hard reload (ignore cache)
F11	Full screen
Ctrl + P	Print page
Ctrl + U	View page source

7.8 Worked examples — Internet

Q7.1. WWW was invented by:
(a) Bill Gates (b) Larry Page
(c) **Tim Berners-Lee** (d) Vint Cerf

Solution: Tim Berners-Lee at CERN, 1989 → (c).

Q7.2. `https://` indicates the site uses:
(a) Hidden Text Tracking Protocol
(b) **HyperText Transfer Protocol Secure (TLS)**
(c) HyperText Tunnel Protocol
(d) Hot Text Transfer Protocol

Solution: HTTPS = HTTP over TLS encryption → (b).

Q7.3. Which key combination opens incognito mode in Chrome?
(a) Ctrl + I (b) Ctrl + Shift + I
(c) **Ctrl + Shift + N** (d) Ctrl + N

Solution: Ctrl + Shift + N for incognito; Ctrl + N is plain new window → (c).

Q7.4. A **cookie** in a browser is:
(a) A virus
(b) An image format
(c) **A small text file stored by a website on your device**
(d) A search engine

Solution: (c).

Q7.5. The "default" port for HTTPS is:
(a) 80 (b) **443** (c) 21 (d) 25

Solution: HTTP=80, HTTPS=443, FTP=21, SMTP=25 → (b).

CHAPTER 8 — E-mail & E-banking

8.0 E-mail basics

Email (electronic mail) is the exchange of messages between users via the Internet. Sent in 1971 by **Ray Tomlinson**, who chose **@** as the separator between user-name and host-name. First commercial email service for the public: **Hotmail (1996)**.

8.1 Anatomy of an email address

```
ramesh.kumar@gmail.com
└──────┬───────┘ └───┬───┘
    user-name     domain (mail server)
```

8.2 Parts of an email message

- **To** — primary recipient(s); everyone sees this list.
- **Cc** (Carbon Copy) — secondary recipients; everyone sees.

- **Bcc** (Blind Carbon Copy) — recipients hidden from others.
- **Subject** — one-line topic.
- **Body** — main content.
- **Attachment** — file accompanying the message (image, PDF, doc).
- **Signature** — automatically appended footer (name, contact).

8.3 Email protocols (must memorise)

Protocol	Full form	Port	Purpose
SMTP	Simple Mail Transfer Protocol	**25** (587 encrypted)	**Send** mail (client → server, server → server)
POP3	Post Office Protocol v3	**110** (995 encrypted)	**Download** mail to local device; usually deletes from server
IMAP	Internet Message Access Protocol	**143** (993 encrypted)	**Synchronise** mail; mail stays on server; multi-device
MIME	Multipurpose Internet Mail Extensions	—	Standard for **attachments + non-ASCII text**

Mnemonic: **S**end with **SMTP**; **r**eceive with **POP** or **IMAP**.

8.4 Popular email clients & services

- **Web clients**: Gmail (Google), Outlook.com / Hotmail (Microsoft), Yahoo Mail, ProtonMail, Zoho Mail.
- **Desktop clients**: Microsoft Outlook (part of Office), Mozilla Thunderbird, Apple Mail, Eudora.
- **Business systems**: Microsoft Exchange Server, IBM Notes (Lotus).

Gmail free quota: **15 GB** shared across Gmail + Google Drive + Google Photos.

8.5 Email-related terms

- **Inbox / Sent / Drafts / Spam (Junk) / Trash / Archive** — standard folders.
- **Forward** — send the received email to a new recipient.
- **Reply** — respond to sender only.
- **Reply All** — respond to sender + all recipients in the To/Cc list.
- **Phishing** — fraudulent email pretending to be legit (more in Cyber Security chapter).
- **Spam** — unsolicited bulk email.
- **Whitelist / Blacklist** — allow / block sender lists.

8.6 E-banking (Internet banking) basics

- **NEFT** (National Electronic Funds Transfer) — batched 24×7 (since Dec 2019); no min/max set centrally.
- **RTGS** (Real Time Gross Settlement) — real-time, 24×7 (since Dec 2020); **min ₹2 lakh**, no upper limit.
- **IMPS** (Immediate Payment Service) — 24×7; via NPCI; max ₹5 lakh.
- **UPI** (Unified Payments Interface) — launched **11 April 2016** by NPCI; instant, free, ID-based (`name@bank`); 24×7; per-transaction limit ₹1 lakh (₹5 lakh for some merchant categories).
- **AePS** (Aadhaar enabled Payment System) — biometric authentication via Aadhaar number.
- **BBPS** (Bharat Bill Payment System) — centralised bill payments (electricity, gas, mobile).

8.7 Card networks & instruments

- **Debit card** — debits your bank account immediately.
- **Credit card** — bank loan, repaid monthly.
- **Prepaid card / Wallet** — load money first (Paytm Wallet, Mobikwik).
- **RuPay** — Indian card network (NPCI, 2012).
- **Visa / MasterCard / American Express / Diners Club / Discover** — international networks.
- **ATM** — Automated Teller Machine; first in India: **HSBC, Mumbai, 1987**.
- **PoS** — Point of Sale terminal at merchant; **MPoS** = mobile PoS.

8.8 Security extras for e-banking

- **OTP** (One-Time Password) — sent via SMS / app for second-factor authentication.
- **CVV** (Card Verification Value) — 3-digit (Visa/MC) or 4-digit (Amex) code on back of card; never share.
- **3-D Secure** — additional online auth (Verified by Visa, MasterCard SecureCode).
- **Tokenisation** — your real card number is replaced by a unique device-specific token for storage.
- **Two-factor authentication (2FA)** — something you know + something you have (password + OTP).

8.9 Worked examples — Email & e-banking

Q8.1. Which protocol is used to **send** email?
(a) POP3 (b) IMAP (c) **SMTP** (d) FTP

Solution: SMTP for sending → (c).

Q8.2. Default port for IMAP (unencrypted):
(a) 25 (b) **143** (c) 110 (d) 443

Solution: IMAP = 143; IMAPS = 993 → (b).

Q8.3. Bcc field is used to:
(a) Send to multiple primary recipients
(b) **Send a hidden copy to recipients without others knowing**
(c) Make subject bold
(d) Attach files

Solution: Blind Carbon Copy → (b).

Q8.4. UPI was launched in India by:
(a) RBI (b) SBI
(c) **NPCI** (d) Ministry of Finance

Solution: NPCI launched UPI on 11 April 2016 → (c).

Q8.5. Minimum amount for RTGS transfer:
(a) ₹1 (b) ₹50,000 (c) ₹1 lakh (d) **₹2 lakh**

Solution: RTGS min = ₹2 lakh; no upper limit → (d).

CHAPTER 9 — Networking — Devices & Protocols

9.0 Types of network (by size)

Type	Range	Example
PAN (Personal Area Network)	1-10 m	Bluetooth between phone & headphone
LAN (Local Area Network)	10 m – 1 km	Office Wi-Fi, home network
CAN (Campus Area Network)	1-5 km	University campus
MAN (Metropolitan Area Network)	5-50 km	City-wide cable TV
WAN (Wide Area Network)	100 km+	The Internet itself
GAN (Global Area Network)	World-wide	Satellite networks
SAN (Storage Area Network)	Data-centre	High-speed storage
VPN (Virtual Private Network)	Logical (overlay)	Securely access office over Internet

9.1 Network topologies (the "shape" of connections)

Topology	Picture	Pros	Cons
Bus	All on one cable	Cheap, easy	Single break breaks all
Ring	Each connects to two neighbours	Predictable	Single break breaks all
Star	All connect to central hub/switch	Failure of one device doesn't affect others	Central failure = total failure
Mesh	Every device connects to every other	Most fault-tolerant	Expensive cabling (n(n-1)/2 links)
Tree (Hierarchical)	Star of stars	Scalable	Backbone failure
Hybrid	Mix of above	Flexible	Complex

9.2 Networking devices — pick the right tool

Device	OSI Layer	Function
Hub	Physical (Layer 1)	Broadcasts to all ports — obsolete
Repeater	Physical (1)	Amplifies signal over distance
Bridge	Data-Link (2)	Connects two LAN segments, filters by MAC
Switch	Data-Link (2)	Smart hub; forwards by MAC; uses CAM table
Router	Network (3)	Forwards packets between networks by IP
Gateway	Layer 4-7	Connects networks with different protocols
Modem	Physical	**MOdulator-DEModulator**; analog ↔ digital (DSL, cable)
NIC	Physical	Network Interface Card; gives PC a MAC address
Access Point (AP)	Physical / Data-Link	Wireless extension of wired LAN
Firewall	Multiple	Filters traffic by rules (more in Cyber Security)

9.3 The OSI model (7 layers — memorise!)

```
  7  Application   (HTTP, SMTP, FTP, DNS, DHCP)
  6  Presentation  (Encryption, compression, JPEG, MP3)
  5  Session       (NetBIOS, RPC, sockets)
  4  Transport     (TCP, UDP) — segments / datagrams
  3  Network       (IP, ICMP) — packets, routers
  2  Data-Link     (Ethernet, Wi-Fi, MAC) — frames, switches
  1  Physical      (cables, hubs, signals) — bits
```

Mnemonic (top-down): **All People Seem To Need Data Processing**. Mnemonic (bottom-up): **Please Do Not Throw Sausage Pizza Away**.

KEY FACT

Mnemonic — OSI layers (bottom-up, Physical to Application): "**P**lease **D**o **N**ot **T**hrow **S**ausage **P**izza **A**way" = **P**hysical → **D**ata Link → **N**etwork → **T**ransport → **S**ession → **P**resentation → **A**pplication. Examiners ask: "Which layer does a Router work at?" → Layer 3 (Network). "Which layer does a Switch work at?" → Layer 2 (Data Link). "Which layer does a Hub work at?" → Layer 1 (Physical).

EXAM ALERT

⚠️ **TRAP — Hub vs Switch vs Router layers:** Hub = Layer 1 (Physical, dumb repeater). Switch = Layer 2 (Data Link, uses MAC addresses). Router = Layer 3 (Network, uses IP addresses). A common trap option says "Switch operates at Layer 3" — only Layer 3 switches (expensive enterprise gear) do; standard switches are Layer 2.

The simpler **TCP/IP model** has 4 layers: Application → Transport → Internet → Network Access.

9.4 Common protocols (must memorise — 1-2 Qs/paper)

Protocol	Full form	Layer	Port
HTTP	HyperText Transfer Protocol	App	80
HTTPS	HTTP Secure	App	443
FTP	File Transfer Protocol	App	20 (data), 21 (control)
SFTP	SSH File Transfer Protocol	App	22
TFTP	Trivial FTP	App	69
SMTP	Simple Mail Transfer	App	25 / 587
POP3	Post Office Protocol 3	App	110 / 995
IMAP	Internet Message Access	App	143 / 993
DNS	Domain Name System	App	53
DHCP	Dynamic Host Configuration	App	67 (server), 68 (client)
SSH	Secure Shell	App	22
Telnet	Terminal emulation (unsafe — no encryption)	App	23
SNMP	Simple Network Management	App	161
LDAP	Lightweight Directory Access	App	389
NTP	Network Time Protocol	App	123

Protocol	Full form	Layer	Port
TCP	Transmission Control Protocol	Transport	—
UDP	User Datagram Protocol	Transport	—
IP	Internet Protocol	Network	—
ICMP	Internet Control Message (ping)	Network	—
ARP	Address Resolution Protocol	Network/Link	—

TCP vs UDP — the only "comparison" you must know:

Feature	TCP	UDP
Connection	Connection-oriented (handshake)	Connection-less
Reliable	Yes (acknowledgements, re-tx)	No
Order	Guaranteed	Not guaranteed
Speed	Slower	Faster
Header size	20 bytes	8 bytes
Used for	Web (HTTP), email, file transfer	Video / voice calls, DNS, gaming, streaming

9.5 IP addressing basics

IPv4 — 32-bit address written as four decimal octets: `192.168.1.10`. Total ~4.3 billion addresses (almost exhausted).

IPv6 — 128-bit address written in 8 hexadecimal groups: `2001:0db8:85a3::8a2e:0370:7334`. Practically unlimited.

Private IP ranges (RFC 1918) — used inside LANs, not routable on Internet: - 10.0.0.0 – 10.255.255.255 (Class A) - 172.16.0.0 – 172.31.255.255 (Class B) - 192.168.0.0 – 192.168.255.255 (Class C)

Special addresses: - `127.0.0.1` — **loopback** (your own machine; "localhost"). - `0.0.0.0` — any address / unspecified. - `255.255.255.255` — broadcast.

Subnet mask — defines which bits are network vs host (e.g., `255.255.255.0` = first 24 bits network, last 8 host).

MAC address — 48-bit physical address (12 hex digits, six pairs separated by `:` or `-`), e.g., `00:1A:2B:3C:4D:5E`. Hard-burned on the NIC. Unique to each device.

NAT (Network Address Translation) — router translates between many private IPs and one public IP.

DHCP — automatically assigns IP, subnet, gateway, DNS to clients on a network.

DNS — translates domain names (google.com) into IP addresses (142.250.183.142). The Internet's "phone book". Hierarchy: Root → TLD (.com) → Authoritative servers.

9.6 Wi-Fi standards (1 Q every few papers)

Standard	IEEE	Marketing name	Max speed
802.11a	1999	—	54 Mbps (5 GHz)
802.11b	1999	—	11 Mbps (2.4 GHz)
802.11g	2003	—	54 Mbps (2.4 GHz)
802.11n	2009	**Wi-Fi 4**	600 Mbps
802.11ac	2013	**Wi-Fi 5**	6.9 Gbps
802.11ax	2019	**Wi-Fi 6**	9.6 Gbps

Standard	IEEE	Marketing name	Max speed
802.11ax-2021	2021	**Wi-Fi 6E**	+ 6 GHz band
802.11be	2024	**Wi-Fi 7**	46 Gbps

Bluetooth — short-range PAN; current = Bluetooth 5.4; range 1-100 m depending on class.

9.7 Mobile data evolution

Generation	Speed	Year (India)	Key feature
1G	2 kbps	1980s	Analog voice only
2G	64 kbps	1995	Digital voice + SMS (GSM)
3G	2 Mbps	2008	Internet, video calls (UMTS)
4G LTE	100 Mbps – 1 Gbps	2012	High-speed Internet, HD video
5G	1-10 Gbps	**Oct 2022 (India)**	Ultra-low latency, IoT, edge computing
6G	100+ Gbps (target)	~2030	THz frequencies, AI integration

9.8 Worked examples — Networking

Q9.1. LAN stands for:
(a) Large Area Network (b) Local Access Network
(c) **Local Area Network** (d) Limited Area Network

Solution: (c).

Q9.2. A **switch** operates at which OSI layer?
(a) Physical (b) **Data-Link** (c) Network (d) Transport

Solution: Switches forward by MAC (Layer 2) → (b).

Q9.3. The loopback IP address is:
(a) 0.0.0.0 (b) 255.255.255.255
(c) **127.0.0.1** (d) 192.168.0.1

Solution: (c).

Q9.4. Which protocol translates domain names to IP addresses?
(a) DHCP (b) HTTP (c) **DNS** (d) FTP

Solution: (c).

Q9.5. Which is **connection-less**?
(a) **UDP** (b) TCP (c) HTTP (d) SMTP

Solution: UDP is connection-less → (a).

Q9.6. IPv6 address length is:
(a) 32 bits (b) 64 bits (c) **128 bits** (d) 256 bits

Solution: IPv6 = 128 bits → (c).

Q9.7. Which device modulates digital signals into analog for transmission over phone lines?
(a) Router (b) Switch (c) **Modem** (d) Hub

Solution: Modem = MOdulator-DEModulator → (c).

Q9.8. OSI model has how many layers?
(a) 4 (b) 5 (c) 6 (d) **7**

Solution: OSI = 7 layers; TCP/IP = 4 → (d).

CHAPTER 10 — Cyber Security — Threats & Prevention

10.0 Why cyber-security is in the syllabus

Government computers handle sensitive data — Aadhaar, tax, defence. A single phishing click can compromise a department. SSC tests basic threat recognition + prevention so that government employees can be the first line of defence.

10.1 The CIA triad — the foundation

Every security policy aims at three properties:

```
                ┌───────────────┐
                │CONFIDENTIALITY│  Only authorised users can READ
                └───────────────┘
                        ▲
                        │
               ┌────────┴────────┐
               │                 │
               ▼                 ▼
     ┌────────────┐     ┌──────────────┐
     │ INTEGRITY  │     │ AVAILABILITY │
     │ Data is    │     │ System is up │
     │ unchanged  │     │ when needed  │
     └────────────┘     └──────────────┘
```

Memory peg: **CIA** = Confidentiality, Integrity, Availability.

10.2 Threat taxonomy — every type explained

A. Malware — software designed to harm

Type	Behaviour	Example
Virus	Attaches to a host file/program; activates when host runs; spreads when shared	ILOVEYOU (2000)
Worm	**Self-propagating** across networks without user action	Stuxnet, WannaCry
Trojan Horse	Disguised as legit software; opens backdoor	Zeus, Emotet
Spyware	Secretly monitors user (keystrokes, browsing)	Pegasus, Cookiethief
Adware	Bombards user with ads	Fireball
Ransomware	Encrypts files; demands ransom	WannaCry (2017), Petya, LockBit
Rootkit	Hides deep in OS; grants persistent admin access	Sony BMG rootkit
Keylogger	Records every keystroke	Refog, RemoteSpy
Bot / Botnet	Compromised devices controlled remotely (zombies)	Mirai (DDoS, 2016)
Logic bomb	Code that triggers on a condition (date, action)	Insider attacks

Type	Behaviour	Example
Cryptojacker	Mines cryptocurrency using victim's CPU	Coinhive
Fileless malware	Lives only in RAM; uses legit tools (PowerShell, WMI)	Astaroth

B. Network attacks

Attack	Description
Phishing	Fake email/SMS lures user into giving credentials. **Spear-phishing** = targeted; **Whaling** = CEO-level.
Vishing	Phishing over voice call
Smishing	Phishing via SMS
DoS (Denial of Service)	Flood a server with requests till it crashes
DDoS (Distributed DoS)	DoS launched from thousands of botnet devices
Man-in-the-Middle (MITM)	Attacker secretly relays/alters communication between two parties
Eavesdropping / Sniffing	Passive interception of network traffic
DNS spoofing	Inject false DNS records to redirect to malicious site
SQL Injection	Insert SQL into web form to leak/modify database
XSS (Cross-Site Scripting)	Inject script into a webpage that other users will run
CSRF (Cross-Site Request Forgery)	Trick logged-in user into making an unwanted request
Zero-day	Exploit for an unknown / unpatched vulnerability
Brute force	Try every password combination
Dictionary attack	Try common words/passwords
Credential stuffing	Reuse leaked passwords on other sites

C. Social-engineering tactics

- **Pretexting** — invented scenario to obtain info ("This is HR; verify your password").
- **Baiting** — leaving infected USB sticks for someone to plug in.
- **Tailgating / Piggybacking** — following an authorised person through a secured door.
- **Quid Pro Quo** — promise of a benefit ("Free anti-virus") in exchange for action.

10.3 Prevention & countermeasures

Control	What it does
Antivirus / Anti-malware	Scans + removes known malware; signature + heuristic based. Examples: Quick Heal, K7, Bitdefender, Kaspersky, Norton, McAfee, Windows Defender.
Firewall	Filters incoming/outgoing traffic by rules (port, IP, protocol). **Hardware firewall** (router) vs **software firewall** (Windows Firewall). **Next-Gen Firewall (NGFW)** adds deep packet inspection + intrusion prevention.
IDS / IPS	Intrusion Detection / Prevention System — monitors and alerts (IDS) or blocks (IPS) suspicious traffic.
Strong password	Min 12 chars, mix of upper/lower/number/symbol; unique per site. Use a **password manager** (Bitwarden, 1Password, LastPass, KeePass).
MFA / 2FA	Multi-factor: password + OTP / app code / biometric. SSC tip: SMS OTP is weakest; **TOTP apps** (Google Authenticator) better.
Encryption	**Symmetric** (AES, DES) — same key for both sides. **Asymmetric** (RSA, ECC) — public/private key pair. TLS uses both.

Control	What it does
Digital signature	Hash + private-key encryption; proves authenticity + integrity.
Digital certificate	X.509 file issued by CA (e.g., DigiCert, Let's Encrypt) binding a public key to an identity.
Hashing	One-way function (MD5 — broken; SHA-1 — weak; SHA-256, SHA-3 — current). Used for passwords + file integrity.
VPN	Encrypted tunnel between client and server; hides IP + protects data on public Wi-Fi.
Backup (3-2-1 rule)	3 copies of data, on 2 different media, with 1 off-site.
Patching / Updates	Apply OS + software patches immediately.
Least Privilege	Users get only the permissions they need.
Awareness training	Train staff to spot phishing / social engineering.

10.4 Indian cyber-security ecosystem (very high-yield)

Body	Established	Role
CERT-In	**2004**	Computer Emergency Response Team; nodal agency for cyber incidents under MeitY
NCIIPC	2014	National Critical Information Infrastructure Protection Centre — protects "critical" sectors (power, banking, defence)
I4C	2018	Indian Cyber Crime Coordination Centre under MHA; runs cybercrime.gov.in & 1930 helpline
NTRO	2004	National Technical Research Organisation — technical intelligence
DRDO Cyber & AI Lab	—	Defence-grade cyber research
Data Security Council of India (DSCI)	2008	Industry body set up by NASSCOM

Key Acts and policies

- **IT Act, 2000** (amended 2008) — primary law for cyber crimes; Sections 43 (damage), 65 (tampering source code), 66 (computer fraud, identity theft, cheating by impersonation), 67 (obscenity), 67A/B (sexual content), 69 (interception powers), 70 (protected systems), 72 (breach of privacy).
- **Digital Personal Data Protection (DPDP) Act, 2023** — India's GDPR; consent-based personal data processing; sets up Data Protection Board.
- **National Cyber Security Policy, 2013** — first formal cyber policy.
- **Indian Computer Emergency Response Team Rules, 2013** — operational rules for CERT-In.

10.5 Worked examples — Cyber Security

Q10.1. A program that replicates itself across networks **without user action** is a:
(a) Virus (b) Trojan (c) **Worm** (d) Adware

Solution: Worm self-propagates; virus needs a host file → (c).

Q10.2. Phishing typically attempts to:
(a) Slow down the network
(b) **Steal credentials by tricking the user**

(c) Crash a server
(d) Encrypt files for ransom

Solution: Phishing = credential theft via deception → (b).

Q10.3. CERT-In is under which ministry?
(a) Ministry of Home Affairs
(b) **Ministry of Electronics & IT (MeitY)**
(c) Ministry of Defence
(d) Ministry of External Affairs

Solution: CERT-In is under MeitY → (b). I4C is under MHA.

Q10.4. The CIA triad in cyber-security stands for:
(a) Central Intelligence Agency
(b) **Confidentiality, Integrity, Availability**
(c) Cyber, Internet, Application
(d) Code, Information, Authentication

Solution: (b).

Q10.5. Which is an example of a **symmetric** encryption algorithm?
(a) RSA (b) ECC (c) **AES** (d) Diffie-Hellman

Solution: AES = symmetric; RSA/ECC/DH = asymmetric → (c).

Q10.6. WannaCry (2017) was a:
(a) Worm (b) Trojan (c) **Ransomware** (d) Adware

Solution: Ransomware exploited the EternalBlue vulnerability → (c).

Q10.7. Which section of the IT Act 2000 deals with computer-related offences like identity theft and cheating by impersonation?
(a) Section 43 (b) Section 65 (c) **Section 66** (d) Section 67

Solution: Section 66 covers identity theft (66C) + cheating by impersonation (66D) → (c).

Q10.8. A **firewall** is best described as:
(a) An anti-virus
(b) **A network filter that controls incoming/outgoing traffic by rules**
(c) A backup utility
(d) A search engine

Solution: (b).

Q10.9. The national helpline number for reporting cybercrime in India:
(a) 100 (b) 112 (c) **1930** (d) 1800

Solution: Cyber-crime helpline = 1930 → (c).

Q10.10. DDoS stands for:
(a) Digital Denial of Service
(b) **Distributed Denial of Service**
(c) Dynamic Denial of Service
(d) Double Denial of Service

Solution: (b).

CHAPTER 11 — Database Basics

11.0 Why databases matter (and why they appear on the exam)

Government systems run on databases — income-tax records, Aadhaar, IRCTC tickets. SSC CGL Tier-2 tests basic database vocabulary so aspirants have working knowledge of how data is stored and managed. Expect 1-2 Qs on DBMS, SQL commands, and ACID properties.

11.1 DBMS vs RDBMS — the key distinction

Concept	Definition	Example
DBMS (Database Management System)	Software that stores, retrieves, and manages data	File-based systems; early databases
RDBMS (Relational DBMS)	DBMS that organises data in **tables (relations)** linked by keys; uses **SQL**	MySQL, PostgreSQL, Oracle, MS SQL Server, SQLite
Table (Relation)	Data organised in **rows (records/tuples)** and **columns (fields/attributes)**	Employees table: EmpID, Name, Dept, Salary
Schema	Blueprint of the database — the structure of tables and relationships	Created before data is entered

KEY FACT

Key rule: Every RDBMS is a DBMS, but not every DBMS is an RDBMS. An RDBMS organises data into related tables. SQL is the language used to communicate with an RDBMS.

11.2 SQL commands — four categories

SQL (Structured Query Language) commands are grouped into four categories — this is a favourite "match the command to its category" question.

Category	Full name	Commands	What they do
DDL	Data Definition Language	`CREATE`, `DROP`, `ALTER`, `TRUNCATE`, `RENAME`	Define or change the **structure** of tables
DML	Data Manipulation Language	`SELECT`, `INSERT`, `UPDATE`, `DELETE`	Work with **data inside** tables
DCL	Data Control Language	`GRANT`, `REVOKE`	Control **access permissions**
TCL	Transaction Control Language	`COMMIT`, `ROLLBACK`, `SAVEPOINT`	Manage **transactions** (ensure ACID)

KEY FACT

Mnemonic — DDL vs DML: DDL = **D**efine the **D**rawing board (structure of tables); DML = **D**ata **M**anipulation (actual data inside). The most tested pair: DDL has `CREATE` and `DROP`; DML has `SELECT` and `INSERT`.

EXAM ALERT

⚠️ **TRAP — TRUNCATE is DDL, not DML:** Many students assume TRUNCATE (which removes all rows) is DML like DELETE. It is NOT — TRUNCATE is a DDL command (it resets the table structure's data; cannot be rolled back in most databases). DELETE is DML and can be rolled back.

11.3 Keys and constraints

Key/Constraint	Definition	Example
Primary Key	Uniquely identifies each row; cannot be NULL; only ONE per table	EmpID in Employees table
Foreign Key	A column in Table B that references the Primary Key of Table A; links two tables	DeptID in Employees referencing DeptID in Departments
Unique Key	All values must be unique, but CAN be NULL; multiple allowed per table	Email column
Candidate Key	Any column (or combination) that can uniquely identify a row	EmpID or SSN — either could be the primary key
Composite Key	Primary key made of two or more columns	(OrderID, ProductID) together
Index	Speeds up search/retrieval on a column; like a book's index	Index on LastName for faster searches
NOT NULL	Column cannot have empty values	
CHECK	Values must satisfy a condition	Age ≥ 18
DEFAULT	Value used if no value provided	Default City = 'Delhi'

11.4 ACID Properties — the most tested concept

ACID is the set of four properties that guarantee database transactions are processed reliably.

```
A — ATOMICITY:   "All or nothing"  — a transaction completes fully, or not at all
                  (if a bank transfer fails mid-way, the debit is reversed)

C — CONSISTENCY: "Always valid"    — a transaction brings the database from one
                  valid state to another (total money before and after transfer = same)

I — ISOLATION:   "Independent"     — concurrent transactions don't interfere with
                  each other (two users booking the last ticket simultaneously)

D — DURABILITY:  "Permanent"       — once committed, data survives crashes/power cuts
                  (committed bank transactions are written to disk, not lost)
```

KEY FACT

Memory peg for ACID: "All Cars In Drive" = Atomicity, Consistency, Isolation, Durability. The exam asks "which property ensures that completed transactions survive a power failure?" → **Durability**.

11.5 Database models (types)

Model	Structure	Example
Relational	Tables + relationships (most used)	MySQL, Oracle, PostgreSQL
Hierarchical	Tree structure (parent-child)	IBM IMS
Network	Graph structure (many-to-many)	IDS system
Object-Oriented	Objects (data + methods together)	ObjectDB
NoSQL	Non-relational; flexible schema	MongoDB (document), Redis (key-value), Cassandra (wide-column), Neo4j (graph)

NoSQL vs SQL — NoSQL databases are used for **big data, real-time web apps, unstructured data** (social media posts, sensor readings). They sacrifice some ACID guarantees for speed and scalability. SQL/RDBMS are used for structured transactional data (banking, ERP).

11.6 Normalisation (quick overview)

Normalisation is the process of organising a database to reduce redundancy (repetition) and improve integrity.

Normal Form	Rule	Common violation it fixes
1NF	Atomic values; no repeating groups	Multiple phone numbers in one cell
2NF	1NF + no partial dependency (all non-key columns depend on the whole primary key)	Data that depends only on part of a composite key
3NF	2NF + no transitive dependency (non-key columns depend only on the primary key, not on each other)	City determined by ZipCode, not directly by the PK
BCNF	Stronger version of 3NF	Uncommon; exam rarely goes beyond 3NF

11.7 Worked examples — Database

Q11.1. Which SQL command is used to retrieve data from a table?

(a) INSERT (b) **SELECT** (c) UPDATE (d) CREATE

Solution: SELECT is the DML retrieval command → (b).

Q11.2. Which is a DDL command?

(a) SELECT (b) INSERT (c) **CREATE** (d) DELETE

Solution: CREATE defines a table structure → (c).

Q11.3. A Foreign Key:

(a) Must be unique in its own table

(b) **References the Primary Key of another table**

(c) Cannot be NULL

(d) Identifies each row uniquely

Solution: (b).

Q11.4. Which ACID property ensures that a committed transaction survives a system crash?

(a) Atomicity (b) Consistency (c) Isolation (d) **Durability**

Solution: (d).

Q11.5. RDBMS stands for:
(a) Rapid Database Management System
(b) **Relational Database Management System**
(c) Remote Database Management System
(d) Redundant Database Management System

Solution: (b).

Q11.6. The command to remove ALL rows from a table without removing the table structure is:
(a) DELETE (b) DROP (c) **TRUNCATE** (d) REMOVE

Solution: TRUNCATE removes all rows but keeps the table structure (DDL); DROP removes the table itself; DELETE (DML) removes rows but can be rolled back → (c).

Q11.7. Which is an example of a NoSQL database?
(a) MySQL (b) Oracle (c) PostgreSQL (d) **MongoDB**

Solution: MongoDB is a document-based NoSQL database → (d).

APPENDIX A — 100 Must-Know Acronyms

Acronym	Expansion
ALU	Arithmetic Logic Unit
API	Application Programming Interface
APN	Access Point Name
ARP	Address Resolution Protocol
ASCII	American Standard Code for Information Interchange
BIOS	Basic Input Output System
BCC	Blind Carbon Copy
BMP	Bitmap
BPS	Bits Per Second
CAD	Computer-Aided Design
CAM	Computer-Aided Manufacturing
CC	Carbon Copy
CD	Compact Disc
CDMA	Code Division Multiple Access
CERT	Computer Emergency Response Team
CGI	Computer Generated Imagery
CMOS	Complementary Metal-Oxide Semiconductor
CPU	Central Processing Unit
CRT	Cathode Ray Tube
CSS	Cascading Style Sheets
CSV	Comma Separated Values
CU	Control Unit

Acronym	Expansion
DBMS	Database Management System
DDR	Double Data Rate
DHCP	Dynamic Host Configuration Protocol
DNS	Domain Name System
DOS	Disk Operating System / Denial of Service
DPI	Dots Per Inch
DRAM	Dynamic RAM
DSL	Digital Subscriber Line
DTP	Desktop Publishing
DVD	Digital Versatile Disc
EEPROM	Electrically Erasable Programmable Read-Only Memory
ENIAC	Electronic Numerical Integrator And Computer
FAT	File Allocation Table
FAX	Facsimile
FDD	Floppy Disk Drive
FORTRAN	Formula Translation
FTP	File Transfer Protocol
GB	Giga Byte
GIF	Graphics Interchange Format
GPRS	General Packet Radio Service
GPS	Global Positioning System
GPU	Graphics Processing Unit
GSM	Global System for Mobile Communications
GUI	Graphical User Interface
HDD	Hard Disk Drive
HDMI	High-Definition Multimedia Interface
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
HTTPS	HTTP Secure
IBM	International Business Machines
IC	Integrated Circuit
ICMP	Internet Control Message Protocol
IDE	Integrated Development Environment / Integrated Drive Electronics
IMAP	Internet Message Access Protocol
IMEI	International Mobile Equipment Identity
IoT	Internet of Things
IP	Internet Protocol
ISDN	Integrated Services Digital Network
ISP	Internet Service Provider
JPEG	Joint Photographic Experts Group
JS	JavaScript

Acronym	Expansion
KB	Kilo Byte
LAN	Local Area Network
LCD	Liquid Crystal Display
LED	Light Emitting Diode
LISP	List Processing
MAC	Media Access Control
MAN	Metropolitan Area Network
MBR	Master Boot Record / Memory Buffer Register
MICR	Magnetic Ink Character Recognition
MIDI	Musical Instrument Digital Interface
MIME	Multipurpose Internet Mail Extensions
MIPS	Million Instructions Per Second
MODEM	Modulator-Demodulator
MP3	MPEG Audio Layer III
MPEG	Moving Picture Experts Group
NIC	Network Interface Card
NTFS	New Technology File System
OCR	Optical Character Recognition
OMR	Optical Mark Recognition
OS	Operating System
OSI	Open Systems Interconnection
PAN	Personal Area Network
PC	Personal Computer / Program Counter
PDA	Personal Digital Assistant
PDF	Portable Document Format
PHP	PHP Hypertext Preprocessor
PNG	Portable Network Graphics
POP	Post Office Protocol
POST	Power On Self Test
PROM	Programmable Read-Only Memory
RAM	Random Access Memory
RDBMS	Relational Database Management System
RGB	Red Green Blue
ROM	Read-Only Memory
RTF	Rich Text Format
SaaS	Software as a Service
SAN	Storage Area Network
SATA	Serial Advanced Technology Attachment
SCSI	Small Computer System Interface
SDLC	Software Development Life Cycle
SDRAM	Synchronous DRAM

Acronym	Expansion
SIM	Subscriber Identity Module
SMTP	Simple Mail Transfer Protocol
SQL	Structured Query Language
SRAM	Static RAM
SSD	Solid State Drive
SSL/TLS	Secure Sockets Layer / Transport Layer Security
TB	Tera Byte
TCP	Transmission Control Protocol
UDP	User Datagram Protocol
UPS	Uninterruptible Power Supply
URL	Uniform Resource Locator
USB	Universal Serial Bus
UTF	Unicode Transformation Format
VGA	Video Graphics Array
VLAN	Virtual LAN
VLSI	Very Large Scale Integration
VoIP	Voice over IP
VPN	Virtual Private Network
WAN	Wide Area Network
WAP	Wireless Application Protocol
Wi-Fi	Wireless Fidelity
WLAN	Wireless LAN
WORM	Write Once Read Many
WWW	World Wide Web
XML	Extensible Markup Language
ZIP	Zigzag In-line Package (also archive format)

APPENDIX B — Full Shortcut Sheet

B.1 Windows OS — top 30

(See Chapter 3.4.)

B.2 MS Word — top 30

(See Chapter 4.4.)

B.3 MS Excel — top 35

(See Chapter 5.6.)

B.4 MS PowerPoint — top 15

(See Chapter 6.3.)

B.5 Browser — top 15

(See Chapter 7.7.)

APPENDIX C — 50-Q Mock with full solutions

(Read each Q for 30 sec, attempt, then check the worked solution.)

Q1. Which is the first generation of computers?
(a) Transistor (b) IC
(c) **Vacuum tube** (d) Microprocessor

Sol: 1st gen used vacuum tubes (1940-56) → (c).

Q2. 1 KB = ?
(a) 1000 bytes (b) **1024 bytes** (c) 8 bytes (d) 1024 bits

Sol: (b).

Q3. Which is volatile memory?
(a) ROM (b) Hard disk (c) **RAM** (d) Flash drive

Sol: (c).

Q4. Default extension of MS Word 2019:
(a) .doc (b) **.docx** (c) .txt (d) .rtf

Sol: (b).

Q5. Shortcut for copy:
(a) **Ctrl + C** (b) Ctrl + X (c) Ctrl + V (d) Ctrl + Z

Sol: (a).

Q6. HDMI carries:
(a) Power only (b) **Both audio + video**
(c) Video only (d) Audio only

Sol: (b).

Q7. A Worm differs from a Virus because it:
(a) Encrypts files
(b) Shows ads
(c) **Self-propagates without a host file**
(d) Mines crypto

Sol: (c).

Q8. A laser printer is:
(a) Impact (b) **Non-impact** (c) Plotter (d) Inkjet

Sol: (b).

Q9. Which device works at Layer 3 (Network) of OSI?
(a) Switch (b) Hub (c) **Router** (d) Bridge

Sol: Routers forward by IP (Layer 3) → (c).

Q10. Maximum size of a single file on FAT32:
(a) 2 GB (b) **4 GB** (c) 8 GB (d) Unlimited

Sol: (b).

Q11. RAM stands for:
(a) Read Access Memory (b) **Random Access Memory**
(c) Rapid Active Memory (d) Real Address Memory

Sol: (b).

Q12. Which shortcut in Excel inserts today's date?
(a) Ctrl + D (b) **Ctrl + ;**
(c) Ctrl + T (d) Ctrl + Shift + D

Sol: (b).

Q13. Cell address $A1 is:
(a) Relative (b) **Absolute** (c) Mixed (d) Range

Sol: (b).

Q14. Which OSI layer handles encryption?
(a) Application (b) **Presentation**
(c) Session (d) Transport

Sol: (b).

Q15. Bcc in email means:
(a) Big Copy Carry (b) **Blind Carbon Copy**
(c) Black Copy Carbon (d) Bcc has no standard meaning

Sol: (b).

Q16. IPv4 address length:
(a) 16 bits (b) **32 bits** (c) 64 bits (d) 128 bits

Sol: (b).

Q17. Which is an example of asymmetric encryption?
(a) AES (b) DES (c) **RSA** (d) Twofish

Sol: (c).

Q18. A pen-drive's memory is:
(a) DRAM (b) SRAM
(c) **Flash (EEPROM)** (d) Magnetic tape

Sol: (c).

Q19. Default port for HTTPS:
(a) 21 (b) 80 (c) 110 (d) **443**

Sol: (d).

Q20. Which Excel formula computes `5 mod 3`?
(a) =MOD(5,3) (b) =5%3 (c) =5 MOD 3 (d) **=MOD(5,3)**

Sol: `=MOD(5,3)` → 2. Answer (a)/(d).

Q21. Which is NOT an output device?
(a) Monitor (b) Speaker (c) Printer (d) **Scanner**

Sol: Scanner is input → (d).

Q22. F1 key generally opens:
(a) File menu (b) Find (c) **Help** (d) Save

Sol: (c).

Q23. Storage capacity of a standard single-layer DVD:
(a) 700 MB (b) **4.7 GB** (c) 8.5 GB (d) 25 GB

Sol: (b).

Q24. Ransomware:
(a) Sends spam
(b) Steals card numbers
(c) **Encrypts files and demands payment**
(d) Mines crypto

Sol: (c).

Q25. Which combo opens Task Manager directly?
(a) Ctrl + Alt + Del (b) Win + Tab
(c) **Ctrl + Shift + Esc** (d) Alt + F4

Sol: (c) (Ctrl+Alt+Del opens the security screen, not Task Manager).

Q26. WWW was invented by:
(a) Bill Gates (b) Steve Jobs
(c) **Tim Berners-Lee** (d) Mark Zuckerberg

Sol: (c).

Q27. Power-On Self-Test runs before:
(a) Shutting down (b) Saving a file
(c) **Loading the OS** (d) Connecting to network

Sol: (c).

Q28. Which protocol assigns IP addresses dynamically?
(a) DNS (b) **DHCP** (c) ARP (d) ICMP

Sol: (b).

Q29. Topology where every device connects to a central switch is:
(a) Bus (b) Ring (c) **Star** (d) Mesh

Sol: (c).

Q30. Default Excel cell pointer movement on pressing Enter:
(a) Stays (b) Moves left (c) **Moves down** (d) Moves right

Sol: (c) by default (configurable in Options).

Q31. PowerPoint shortcut to start show from current slide:
(a) F5 (b) Ctrl + F5 (c) **Shift + F5** (d) Alt + F5

Sol: (c).

Q32. Which is the **fastest** memory?
(a) HDD (b) RAM (c) **Register** (d) Cache

Sol: Registers are inside CPU, fastest → (c).

Q33. Largest unit (smallest in this list is "byte"):
(a) **PB** (b) TB (c) GB (d) MB

Sol: PB = Petabyte → (a).

Q34. A digital signature provides:
(a) Confidentiality only (b) **Integrity + Authenticity**
(c) Availability (d) Encryption only

Sol: Hash + private-key signing → (b).

Q35. UPI was launched by:
(a) RBI (b) **NPCI** (c) Visa (d) RuPay

Sol: (b).

Q36. Which storage type has NO moving parts?
(a) HDD (b) DVD drive (c) Floppy (d) **SSD**

Sol: (d).

Q37. Which key combo locks Windows?
(a) Win + D (b) **Win + L** (c) Alt + L (d) Ctrl + L

Sol: (b).

Q38. Phishing typically arrives via:
(a) USB drive (b) **Email or SMS**
(c) Bluetooth (d) Wi-Fi packet

Sol: (b).

Q39. In Excel, F4 inside a formula:
(a) Recalculates the sheet (b) Inserts a chart
(c) **Toggles reference type** (d) Closes the workbook

Sol: (c).

Q40. Which is a **client-side** scripting language?
(a) PHP (b) Python (c) **JavaScript** (d) SQL

Sol: (c) (JS runs in the browser).

Q41. Mouse was invented by:
(a) Bill Gates (b) Steve Jobs
(c) **Douglas Engelbart** (d) Alan Turing

Sol: (c), 1964.

Q42. Which is a **search operator** in Google?
(a) `+include` (b) `site:`
(c) `filetype:` (d) **All of the above**

Sol: All work → (d).

Q43. NTFS supports:
(a) Max 4 GB file
(b) **Files larger than 4 GB + permissions + journaling**
(c) Linux only
(d) Not journaled

Sol: (b).

Q44. Which OSI layer handles IP addresses?
(a) Physical (b) Data-Link (c) **Network** (d) Transport

Sol: (c).

Q45. Which is a non-impact printer?
(a) Daisy wheel (b) Dot matrix (c) Drum (d) **Inkjet**

Sol: (d) (laser & inkjet are non-impact).

Q46. First commercial email service for the public:
(a) Gmail (b) Yahoo Mail (c) **Hotmail** (d) Outlook

Sol: Hotmail launched 1996 → (c).

Q47. Default port of FTP control connection:
(a) 20 (b) **21** (c) 22 (d) 23

Sol: Data = 20, Control = 21 → (b).

Q48. Which is a **distributed** OS?
(a) DOS (b) Windows 7 (c) **Plan 9** (d) Mac OS Classic

Sol: (c).

Q49. WannaCry exploited which vulnerability family?
(a) Heartbleed (b) Spectre
(c) **EternalBlue (SMBv1)** (d) Shellshock

Sol: (c).

Q50. Cybercrime national helpline in India:
(a) 100 (b) 112 (c) 181 (d) **1930**

Sol: (d).

APPENDIX D — 7-day final revision plan

Day	Focus	Hours
1	Chapter 1 (Hardware) + Chapter 2 (Memory)	3
2	Chapter 5 (Excel) — start (functions + shortcuts)	3
3	Chapter 5 (Excel) — finish + Chapter 4 (Word)	3
4	Chapter 3 (Windows OS) + Chapter 6 (PowerPoint)	2
5	Chapter 9 (Networking — protocols + devices)	3
6	Chapter 10 (Cyber Security) + Chapter 11 (Database Basics) + Chapter 7-8 (Internet + Email)	3
7	Appendix C 50-Q mock + Appendix A acronym drill	3

Day-before-exam ritual: Re-read Appendix A (acronyms) once + flip through every Shortcut table (B.1 to B.5) once + skim the CIA-triad and protocol-port table. Sleep.

Exam-hall protocol: 1. The first scan should target Excel + Hardware Qs first (highest accuracy). 2. Skip any obscure file-extension Q if you're unsure — guessing is dangerous because all four options look alike. 3. Cyber-security Qs reward common-sense — pick the safest-sounding option if you don't know exactly.

> ***You only need 24 / 60 (8 of 20 Qs)*** *to clear the qualifying cut-off. With this book you should target* ***15 / 20*** *comfortably.*

Final words

Computer Knowledge is **the easiest section to score in** if you simply read the material. The vocabulary is intimidating only the first time; on second exposure each acronym becomes familiar. Use this book for 40 hours over 7-10 days and you will outscore 90 % of candidates on this section.

Good luck.